SPPU - BE

TE - AIDS

CS - Unit I and Unit II

Question 1 (Marks: 7.5)

* Sub-question 1.1 (Marks: 3.0) (Taxonomy Level : Understand)

  Explain the operational model of network security.

* Sub-question 1.2 (Marks: 2.0) (Taxonomy Level : Remember)

  Define vulnerability and write three examples.

* Sub-question 1.3 (Marks: 2.5) (Taxonomy Level : Apply)

  Classify the different categories of information security techniques.

Question 2 (Marks: 7.5)

* Sub-question 2.1 (Marks: 2.0) (Taxonomy Level : Remember)

  Compare symmetric and asymmetric encryption methods.

* Sub-question 2.2 (Marks: 3.5) (Taxonomy Level : Apply)

  Describe the operation of Feistal cipher and explain its role in block ciphers.

* Sub-question 2.3 (Marks: 2.0) (Taxonomy Level : Understand)

  Explain the concept of weak keys in DES and how they can be exploited.

Question 3 (Marks: 7.5)

* Sub-question 3.1 (Marks: 3.0) (Taxonomy Level : Analyze)

  Discuss the differences between security and privacy.

* Sub-question 3.2 (Marks: 2.5) (Taxonomy Level : Evaluate)

  Evaluate the strengths and weaknesses of AES compared to DES.

* Sub-question 3.3 (Marks: 2.0) (Taxonomy Level : Create)

  Develop a simple transposition cipher and demonstrate its encryption and decryption mechanisms.

Question 4 (Marks: 7.5)

* Sub-question 4.1 (Marks: 2.5) (Taxonomy Level : Remember)

  Explain the basic terminology in network security, such as firewall, IDS, and VPN.

* Sub-question 4.2 (Marks: 3.0) (Taxonomy Level : Apply)

  Describe the steps involved in developing a security policy for an organization.

* Sub-question 4.3 (Marks: 2.0) (Taxonomy Level : Understand)

  Discuss the importance of regular security audits and their role in improving network security.